Mixed involutions in orthogonal strings of dimension five

Human Compression project: F001 working proof

Unreviewed working manuscript

This draft completes a proposed route from the accepted bound for negative reflections to a bound for arbitrary involution types. It has not received a complete mathematical review. In particular no principal F001 recovery is inferred from its presence in the working directory.

Let V be a nondegenerate orthogonal five-space in odd characteristic, with symmetric form B. A *negative reflection* is $-r_v$, where $r_v(x) = x - 2B(x,v)v/B(v,v)$ and $B(v,v) \neq 0$. Its negative eigenspace has dimension four. Every other nonidentity involution in $\mathrm{SO}(V)$ has a two-dimensional negative eigenspace; we call it a halfturn. Throughout the endpoint argument we extend scalars to an algebraic closure.

1 The end pairs

Lemma 1. *Suppose $s_0, \dots, s_{r-1}$, $r \geq 6$, is a connected string of involutions in $\mathrm{SO}(V)$ generating an absolutely irreducible group. Then $s_0, s_1, s_{r-2}, s_{r-1}$ are negative reflections.*

Proof. Put $A = \langle a, b\rangle = \langle s_0, s_1\rangle$ and $D = \langle s_{r-2}, s_{r-1}\rangle$. Both are nonabelian dihedral groups and commute elementwise. Every mark lies in $A, D, C(A)$ or $C(D)$. Thus a proper subspace preserved by these four groups contradicts irreducibility. Also if $C(A)$ and $C(D)$ commute elementwise, then $s_2 \in C(D)$ commutes with $s_3 \in C(A)$, another contradiction. We show that a, b must be negative reflections. Reversing the string then gives the other two marks.

Set $h = ab$. It has determinant one, $h^2 \neq 1$, and $aha = h^{-1}$. Its orthogonal primary spaces pair only reciprocal eigenvalues. Consider first semisimple h. If its nonunit reciprocal eigenvalues are all simple and occupy four dimensions, its orthogonal centralizer is a torus times signs and is abelian, which cannot contain D. If a nonunit reciprocal pair has multiplicity two, let E, E' be its two eigenspaces. Choose a basis of E and its image under a as basis of E'. On $E \oplus E'$ the involution a exchanges the planes and the form has matrix

$$\begin{pmatrix} 0 & S \\ S & 0 \end{pmatrix}, \qquad S = S^{\mathsf{T}} \text{ invertible.}$$

The remaining line is fixed by a, by determinant one. Thus $C_{\mathrm{SO}(V)}(A)$ consists of $\mathrm{diag}(T, T, 1)$ with $T^{\mathsf{T}}ST = S$. The noncommuting involutions of D act on this two-dimensional multiplicity space as noncommuting orthogonal reflections; their action is irreducible. The common fixed space of D is therefore exactly the remaining line, since its action on the four-space is two copies of that irreducible two-space. Its centralizer preserves this line, which is already fixed by $A, D, C(A)$. This contradicts irreducibility.

The remaining semisimple case has one simple nonunit reciprocal pair on a plane X and a three-dimensional unit-primary complement. On X, a is a reflection, so it has determinant -1 there. If h is the identity on the complement, a has either one or three negative eigenvalues on that complement. With three, a, b are the desired negative reflections. With one, the complement splits into a positive plane Y and a negative line. The full centralizer of A preserves Y and acts on it as $\mathrm{O}(Y)$, while acting by signs on the other constituents. Hence a nonabelian D acts irreducibly on Y and by scalar characters elsewhere. Its centralizer also preserves Y. This is a

common proper invariant plane. If instead h is negative on a plane Y and positive on a line, commuting with h already separates these constituents. For $C(A)$ to contain a nonabelian D, a must be scalar on Y; otherwise all its joint eigenspaces there are lines and the centralizer is abelian. Again D acts irreducibly on Y and its centralizer preserves that plane, giving the same contradiction. This exhausts semisimple h, since a spectrum consisting only of $1, -1$ would give $h^2 = 1$.

For completeness we now treat every nonsemisimple primary possibility. An even Jordan block at eigenvalue 1 or -1 occurs with even multiplicity in an orthogonal space of odd characteristic. One sees this directly on the leading pairings between equal-sized Jordan chains: the induced nondegenerate form on their multiplicity space is alternating when the chain length is even. Hence that multiplicity is even. This usual orthogonal Jordan restriction leaves in dimension five only the partitions (5), $(3,1,1)$ and $(2,2,1)$ when the whole spectrum is positive unit. The nontrivial negative-primary possibilities are $(3,1)$ or $(2,2)$, with a positive residual line; or a negative scalar plane together with a positive three-chain. A nonunit reciprocal pair with two size-two blocks has abelian centralizer: its action is determined on one block by a polynomial in that Jordan block. A simple nonunit pair together with a positive three-chain has a torus times the polynomial centralizer of the three-chain, again abelian. A regular five-chain also has polynomial, hence abelian, centralizer. None can contain D.

Here are explicit centralizers for all the remaining cases. They also justify that no unipotent mixing of isomorphic constituents has been silently discarded.

For a positive $(2,2,1)$ partition use a basis u_1, u_2, v_1, v_2, z, with $B(u_i, v_j) = \delta_{ij}$, $B(z,z) = 1$, and

$$h = 1 + N, \quad Nv_1 = u_2, \quad Nv_2 = -u_1, \quad Nu_i = Nz = 0.$$

The relation $aNa = -N$ forces determinant -1 on $\operatorname{im} N$: the induced alternating form on this plane is changed to its negative. Choose its two a-eigenlines and isotropic dual vectors in the matching nondegenerate eigenspaces of a. Their residual perpendicular line is fixed by determinant one. Thus $a = \operatorname{diag}(1,-1,1,-1,1)$ in this basis. Solving commutation with N, a, followed by preservation of B, gives exactly the following matrices $M(t,u)$, $t \neq 0$:

$$\begin{aligned} &Mu_1 = tu_1, && Mu_2 = t^{-1}u_2, && Mv_2 = tv_2, \\ &Mv_1 = -\tfrac{1}{2}tu^2u_1 + t^{-1}v_1 + uz, && Mz = -tuu_1 + z. \end{aligned} \tag{1}$$

For clarity, commutation with a separates u_2, v_2 from u_1, v_1, z; commutation with N fixes the two image lines; their dual pairings then determine the displayed coefficients. The possible sign on z is positive because the total determinant is one. Conversely (1) satisfies all the equations. Its multiplication law is

$$M(t,u)M(v,w) = M(tv, w + u/v).$$

Every nonidentity involution has $t = -1$. Two noncommuting involutions have distinct parameters, and their product is $M(1, w-u)$. For $c \neq 0$, the image of $(M(1,c) - 1)^2$ is exactly $\langle u_1 \rangle$, since it maps v_1 to $-c^2u_1$ and kills the other basis vectors. Thus $C(D)$ preserves that line. It is already preserved by $A, D, C(A)$, contradicting irreducibility.

For a positive $(3,1,1)$ partition write $V = W \perp Y$, $W = \langle e, z, f \rangle$, where

$$B(e,f) = 1, \quad B(z,z) = 1, \quad Nz = e, \quad Nf = -z, \quad Ne = 0, \qquad h|_W = 1 + N + N^2/2.$$

The line $\operatorname{im} N^2$ has an a-eigenvalue $\alpha = \pm 1$. Choose f in that eigenspace with $B(e,f) \neq 0$. Its N-chain spans a nondegenerate a-stable three-space, giving the stated perpendicular decomposition after normalization. On e, z, f the signs of a are $\alpha, -\alpha, \alpha$, and its involution a_Y on Y has determinant $-\alpha$. The whole centralizer is

$$\begin{aligned} &Me = ce, && Mz = cz, \\ &Mf = cf + y - \tfrac{B(y,y)}{2c}e, && Mu = Tu - cB(y, Tu)e \quad (u \in Y), \end{aligned} \tag{2}$$

where $c = \pm 1$, $T \in \mathrm{O}(Y)$, $\det T = c$, $Ta_Y = a_Y T$, and $a_Y y = \alpha y$. Indeed commutation with N, a first fixes Me, Mz and the allowed Y component of Mf; the equations $B(Mf, Mf) = 0$ and $B(Mf, Mu) = 0$ determine the remaining terms. They conversely verify (2).

If $\alpha = -1$, then $a_Y = \beta I$. When $\beta = -1$, both a, b are negative reflections, as desired. When $\beta = 1$, $y = 0$; $C(A)$ preserves Y, and a nonabelian D acts irreducibly on that plane and scalarly on W. Therefore $C(D)$ preserves Y too, contradicting irreducibility. If $\alpha = 1$, a_Y is a reflection. Write $T = \mathrm{diag}(\lambda, \mu)$ in its positive and negative lines, and normalize the allowed translation by $u = t/c$. The group law becomes

$$(\lambda_1, \mu_1, u_1)(\lambda_2, \mu_2, u_2) = (\lambda_1\lambda_2, \mu_1\mu_2, u_1 + \mu_1 u_2). \tag{3}$$

This is a central C_2 times an additive line extended by inversion. Noncommuting involutions must both have $\mu = -1$ and distinct translations. Their product has nonzero translation and possibly a central sign; its square is a nonzero pure translation. By (2), its squared nilpotent image is $\langle e \rangle$. This line is preserved by $C(D)$ and already by $A, D, C(A)$, a contradiction.

If the negative primary partition is $(2, 2)$, with a positive line, formula (1) applies but cross-primary mixing is forbidden, so $u = 0$. The centralizer is abelian. For negative partition $(3, 1)$ and a positive line, formula (2) applies with a one-dimensional Y inside the negative primary space and a separate positive line Z. Put $a_Y = \beta$, $a_Z = \gamma$; determinant one gives $\alpha\beta\gamma = -1$. If $\beta \neq \alpha$, translations vanish and the centralizer is abelian. Otherwise (2), with the extra sign on Z fixed by determinant, has exactly the group structure (3). The same characteristic-line argument applies. Finally, for a positive three-chain and a negative scalar plane, cross-primary mixing again forces $y = 0$ in (2). If the restriction of a to the plane is a reflection its centralizer is abelian; if scalar, every nonabelian D and its centralizer preserve that plane. Thus this last case is also reducible. All nonsemisimple cases are exhausted, and the lemma follows. □

2 The middle marks and the length of the string

We now assume the intersection property as well. We acquire from the reviewed rank-contraction paper L037 the following elementary consequence: in a connected C-triple (a, b, c) with $[a, c] = 1$, the element b does not commute with ac. Its complete proof is an actual dependency. We also acquire the reviewed tridiagonal lemma: a connected string of distinct negative reflections in dimension five has length at most six; at length six either end five roots form a basis.

Let X and Y be the root planes of the first and last pairs from Lemma 1. They are perpendicular. If their intersection is nonzero, it is the radical line of both planes, and $A, D, C(A), C(D)$ all preserve it. Thus $X \cap Y = 0$. Their four-dimensional sum has radical of dimension at most one, so at most one plane is degenerate. Reverse the string if necessary so that Y is nondegenerate.

The mark s_2 commutes with both last negative reflections. It therefore acts by one scalar sign on Y, because their nonorthogonal root lines have the same eigenvalue under any commuting orthogonal involution. On $Y^\perp$ it has determinant one, and is a three-dimensional halfturn: identity would make it commute with s_1. Let v be its positive axis there and let v_0 be the first root. Commutation with s_0 gives either $v \perp v_0$ or $v = v_0$. In the latter case either $s_2 = s_0$, or $s_0 s_2$ is a scalar sign on Y and identity on $Y^\perp$, so commutes with s_1. Both alternatives contradict the C-property. Thus $v \perp v_0$. If s_2 is a halfturn on V, it acts negatively on v_0, and $s_0 s_2$ is a negative reflection. Otherwise s_2 already is one. The replacement preserves every string commutation and preserves noncommutation with s_1 by the acquired C-triple fact.

The other neighboring middle mark has the same property. If its commuting root plane X is degenerate, a halfturn commuting with both roots must fix X pointwise: its nondegenerate negative plane cannot equal the degenerate plane X. It must act negatively on the last root v_{r-1}, for otherwise its nondegenerate positive three-space would equal the degenerate span $X + \langle v_{r-1} \rangle$. Multiplying by s_{r-1} again gives a negative reflection.

If $r \geq 8$, the adjacent marks s_3, s_4 lie in $C(A) \cap C(D)$. When both root planes are nondegenerate, this is the abelian group of scalar signs on $X, Y, (X + Y)^\perp$, with the last sign forced

positive by determinant. When one plane is degenerate, the three-chain calculation above shows that the common centralizer is just the scalar sign on the other plane. In either case adjacent commutation is a contradiction. Thus $r \leq 7$.

At $r = 7$, the middle mark s_3 lies in that common centralizer. In the degenerate case it already commutes with s_2, impossible. In the nondegenerate case write $V = X \perp Y \perp Z$. To fail to commute with $s_2 \in C(D)$, s_3 must have negative sign on X; to fail to commute with $s_4 \in C(A)$, it must have negative sign on Y. Its sign on Z is positive, so it is a negative reflection. Replacing halfturns s_2, s_4 by $s_0 s_2, s_4 s_6$ now gives a connected string of seven negative reflections generating the same group. Noncommutation at the inner bonds is unchanged, because the multiplying endpoint commutes with both original marks there. The tridiagonal lemma excludes this. Consequently $r \leq 6$.

At $r = 6$ perform these same replacements, if needed, at positions $2, 3$, using endpoints $0, 5$. Denote the resulting negative reflections by $t_0, \ldots, t_5$ and their roots by $v_0, \ldots, v_5$. They form a connected string generating the same group; either five consecutive roots are a basis. No intersection property for the changed marking is asserted or used.

3 The reflection subgroup of an original end

Put $G = \Omega_5(q)$, with odd $q \geq 5$. We retain its simplicity and absolute irreducibility, and the classical order formula

$$\omega(q) = |G| = \tfrac{1}{2} q^4 (q^2 - 1)(q^4 - 1), \qquad |\mathrm{O}_5(q)| = 4\omega(q). \tag{4}$$

A C-marking of G is connected: disconnected components would give commuting nontrivial normal factors. Suppose that its rank is six and form the negative reflections $t_i = -r_{v_i}$ just obtained. The original end group $H = \langle s_0, \ldots, s_4 \rangle$ is proper, by the intersection property. It contains negative reflections with roots v_0, v_1, v_2, v_4, since t_2 is either s_2 or $s_0 s_2$. If s_3 is negative it contains the fifth root v_3 as well. Otherwise $s_3 = t_3 t_5 = r_{v_3} r_{v_5}$, and conjugating t_2 by s_3 gives a negative reflection with root

$$u = v_2 - c v_3, \qquad c = 2B(v_2, v_3)/B(v_3, v_3) \neq 0. \tag{5}$$

The five roots v_0, v_1, v_2, u, v_4 form a basis. Their pairing graph is connected: v_0, v_1, v_2 form a path, u pairs nontrivially with v_1 and v_4. Therefore, by the acquired invariant-subspace argument for connected spanning roots, the ordinary reflection group K generated by all H-conjugates of these reflections is absolutely irreducible. It is normalized by H, and its signed image

$$\pi(K) \leq H, \qquad \pi(x) = \det(x) x. \tag{6}$$

The kernel of π on the full orthogonal group is $\{1, -1\}$.

We need here the full finite reflection classification, rather than its specialization to five generating reflections. We retain the following dimension-five consequence of Zalesskii–Serezhkin, in the form described by Monson [1, pp.15–18]. Up to extending scalars and conjugating, an absolutely irreducible finite group generated by orthogonal reflections in odd characteristic is one of

1. a group of orthogonal type over a finite field $\mathbf{F}_{q_0}$ of that characteristic, containing $\Omega_5(q_0)$ and preserving a nondegenerate form over $\mathbf{F}_{q_0}$;
2. the faithful ordinary modular reflection representation of A_5, B_5 or D_5;
3. the deleted representation of S_7 in characteristic seven.

For A_5 the characteristic does not divide six. The exceptional five-dimensional quotient of E_6 in characteristic three is already of orthogonal type, $\mathrm{O}_2(5,3)$ in the notation of the notes. The unitary cases in the full classification which admit an invariant nondegenerate symmetric form reduce to the listed orthogonal cases; the notes discuss this immediately after Theorem 4.1. We do not use that theorem's exclusion of cases requiring more than five reflections. The complete classification remains an external retained result.

Lemma 2. *The group K cannot be of orthogonal type.*

Proof. Conjugate so that K and a nondegenerate invariant form B_0 are defined over $\mathbf{F}_{q_0}$. The form B is a scalar multiple of B_0, by absolute irreducibility. We first show that the $\mathbf{F}_{q_0}$-linear span of K is the whole matrix algebra $M_5(\mathbf{F}_{q_0})$. For a reflection with root v over that field the rank-one operator

$$P_v = (1 - r_v)/2 = vB_0(v, -)/B_0(v, v)$$

is in this span, which is an algebra. The graph of all reflecting root lines, joined by nonzero pairings, is connected and spans V; otherwise the spans of its components give a proper invariant space. Products of the P_v along a path are nonzero scalar multiples of $vB_0(w, -)$ for its two end roots. Choosing five root lines as a basis now gives all matrix units in that basis. This proves the claim.

Since H normalizes K, it normalizes this matrix algebra. We give the elementary normalizer argument to avoid an implicit field-automorphism assumption. If h normalizes it, put $F_{ij} = hE_{ij}h^{-1}$, whose entries are in $\mathbf{F}_{q_0}$. Choose a nonzero vector w over that field in the image of F_{11}. The vectors $w_i = F_{i1}w$ form a basis, since $F_{ij}w_k = \delta_{jk}w_i$. Let A have these columns. Then $A \in \mathrm{GL}_5(q_0)$ and $A^{-1}h$ commutes with every E_{ij}, hence equals a scalar c. Thus $h = cA$. The equations preserving B_0 give $c^2 \in \mathbf{F}_{q_0}$, by any nonzero entry of B_0; determinant one gives $c^5 \in \mathbf{F}_{q_0}$. Consequently $c = c^5/(c^2)^2 \in \mathbf{F}_{q_0}$, and $H \leq \mathrm{SO}_5(q_0)$.

The roots v_0, v_1, v_2 are therefore lines defined over $\mathbf{F}_{q_0}$. If s_3 is negative, so is the root v_3, and the missing root line $\langle v_5 \rangle$ is the perpendicular of their four-dimensional span. If s_3 is a halfturn, its negative plane is $\langle v_3, v_5 \rangle$, and

$$\langle v_5 \rangle = \langle v_3, v_5 \rangle \cap \langle v_0, v_1, v_2 \rangle^{\perp}. \tag{7}$$

The intersection has dimension one because v_3 pairs nontrivially with v_2, whereas v_5 pairs with none of the first three roots. Both spaces on the right are defined over $\mathbf{F}_{q_0}$. In either case s_5 is defined over that field too, so $G = \langle H, s_5 \rangle \leq \mathrm{SO}_5(q_0)$ after this conjugation.

By (6), $\omega(q_0) \leq |\pi(K)| \leq \omega(q)$: the subgroup $\Omega_5(q_0)$ has trivial intersection with the kernel of π, since -1 has determinant -1. Thus $q_0 \leq q$. If $q_0 < q$, these are powers of the same odd prime, so $q \geq 3q_0$ and (4) gives $\omega(q) > 4\omega(q_0)$, contradicting the inclusion of G in $\mathrm{O}_5(q_0)$. If $q_0 = q$, (6) shows that $\pi(K) = G \leq H$, contradicting the properness of the end group. □

4 The finite end groups

We next determine the possible original end H, not merely K. In the A_5 case $K \cong S_6$, and in the deleted case $K \cong S_7$. In either representation the ordinary reflections are precisely the transpositions. Indeed an involution with k disjoint transpositions has negative eigenspace dimension k on the deleted or sum-zero module, as the removed trivial constituents are positive. A linear normalizer therefore induces an automorphism preserving the transposition set and its commutation graph. Its vertices are the edges of the complete graph on six or seven letters, with adjacency when they meet. Every family of at least four pairwise intersecting edges is contained in a star: after edges ab, ac are chosen, an edge missing a must be bc, leaving at most that triangle. Thus the maximal cliques of size at least four are exactly the stars. An automorphism permutes these stars, hence permutes the letters, and their pairwise intersections determine every transposition. It follows that the induced group automorphism is inner. After multiplying an element of H by

the appropriate element of $\pi(K)$, it centralizes the absolutely irreducible group K, so is scalar. A scalar in SO_5 is the identity, and consequently $H = \pi(K)$ in both cases.

For B_5 and D_5, the largest normal 2-subgroup of K is the full or even diagonal sign group. Indeed the quotient is S_5, whose largest normal 2-subgroup is trivial. Its five distinct common weight spaces are the coordinate axes, so every element of H permutes those axes. An orthogonal monomial matrix has entries ± 1 in its nonzero positions. The determinant-one monomial group is

$$W = E_5 \rtimes S_5, \qquad E_5 = \{(\epsilon_i) \in \mathbf{F}_2^5 : \textstyle\sum \epsilon_i = 0\}, \tag{8}$$

where a permutation is represented by its signed matrix $\mathrm{sgn}(\sigma)P_\sigma$. Both B_5 and D_5 have signed image equal to W. Thus $H = W$ in these cases.

Lemma 3. *If $H \cong S_6$ or $H = W$ above, every one of its five original marks is a negative reflection.*

Proof. For S_6 we acquire the reviewed classification of independent extremal strings: five involutions generating S_6 independently and satisfying the string commutations form a Coxeter path, up to an automorphism. In the present signed representation a transposition is a negative reflection, whereas an outer image, a triple transposition, is a halfturn. Since s_0, s_1, s_4 are negative reflections, only the first possibility occurs.

For W we acquire the reviewed even-sign module lemma and the extremal independent string result for S_5 from the F006 recovery. Choose a minimal subfamily L whose images generate S_5. The module lemma permits at most one mark outside L, and Whiston's bound gives $|L| \leq 4$. Since all five marks are independent, exactly four belong to L. Their images form a Coxeter path in S_5. Adjacent images do not commute, so their original positions were consecutive as well. The omitted mark x is therefore an endpoint. Reverse the string if needed to put it first.

It commutes with the last three members of L, whose images generate an S_4. The centralizer of this S_4 in S_5 is trivial, so x belongs to E_5. It is fixed by that S_4. The only nonzero even-weight vector constant on its orbits of sizes one and four has weight four. Thus x is a negative reflection.

Put $H_0 = \langle L \rangle$. Its kernel over S_5 is an S_5-submodule of E_5. The acquired module lemma says it is zero or all E_5, as the nonzero constant vector has odd weight. The latter would put x in H_0, contrary to independence. Hence $H_0 \cong S_5$. Each mark in L is consequently the unique lift in H_0 of a transposition. The lifts of permutations of its three fixed coordinates commute with it. They permute those three axes, so the signs of the mark on these axes are equal. Its swapped pair contributes determinant -1, and total determinant one forces each fixed sign to be -1. The swapped pair contributes one negative eigenvalue, giving four in all. Thus these four marks are negative reflections too. □

5 A deleted symmetric end cannot extend

We give the remaining end argument in coordinates, including the quotient by the constant line. Suppose K has the deleted S_7 representation in characteristic seven, so $H = \pi(K) \cong S_7$. In the seven-coordinate permutation space put

$$T = \{(x_i) : \textstyle\sum_{i=1}^{7} x_i = 0\}, \qquad V = T/\langle \mathbf{1} \rangle, \qquad \mathbf{1} = (1, \ldots, 1). \tag{9}$$

The dot product induces a nondegenerate form on V. The negative eigenspace of an involution with k disjoint transpositions has dimension k in V; its determinant is $(-1)^k$. After signing, its negative dimension is four for $k = 1$, two for $k = 2$, and two for $k = 3$. Thus the negative reflections in H are precisely the signed transpositions.

Ignore the scalar signing when discussing the associated permutations. Relabel coordinates so that

$$s_0 = (12), \qquad s_1 = (23), \qquad s_4 = (45). \tag{10}$$

This is possible since the first pair does not commute and the last mark commutes with both. The element s_3 centralizes $S_{\{1,2,3\}}$, so acts only on $\{4,5,6,7\}$ and does not commute with (45). It is therefore a single or double transposition there. The condition that s_2 commutes with both (12) and (45) gives

$$s_2 = (12)^{\epsilon}(45)^{\delta}\tau, \qquad \epsilon, \delta \in \{0,1\}, \quad \tau \in \{1, (36), (37), (67)\}. \tag{11}$$

Transitivity of $H = S_7$ forces $\tau = (36)$ or (37). If s_3 is a single transposition, normalize it to (46). Then $\tau = (36)$ leaves coordinate seven fixed, while $\tau = (37)$ leaves the orbits $\{1,2,3,7\}$ and $\{4,5,6\}$ separate. Neither is transitive. Thus, after relabeling the last four coordinates,

$$s_3 = (46)(57).$$

Conjugating by (45)(67) preserves this and s_4 and exchanges (36) and (37). Hence we may take $\tau = (36)$ in (11).

The negative reflection s_5 commutes with $s_0, \ldots, s_3$, so its root is a common eigenline of these permutations in V. A common eigenline of $(12), (23)$ must be positive for each: neither one-dimensional negative eigenspace is preserved by the other. It therefore has a representative with first three coordinates equal, say $x_1 = x_2 = x_3 = c$. There is no lift ambiguity for a positive involution eigenvector: if $(g-1)x = a\mathbf{1}$, applying g once more gives $2a = 0$, hence $a = 0$. A negative eigenvector modulo the constant line can be made an exact negative eigenvector by subtracting a suitable constant vector.

If the root has negative eigenvalue for the permutation s_2, choose that exact negative representative. Its first three coordinates are then zero, as is x_6. When $\delta = 0$ all coordinates vanish. When $\delta = 1$ the only remaining possibility is a multiple of $e_4 - e_5$, which is not an eigenline of (46)(57), even modulo constants. Thus the root is positive for s_2, and

$$x_6 = c, \qquad \delta = 1 \Longrightarrow x_4 = x_5. \tag{12}$$

If it is also positive for s_3, then $x_4 = x_6 = c$ and $x_5 = x_7 = d$. The sum-zero condition is $5c + 2d = 0$, which in characteristic seven says $d = c$. Its quotient class is zero. Finally suppose it is negative for s_3, and subtract a constant to obtain an exact negative eigenvector. This preserves (12). Since s_3 fixes the first three coordinates, $c = 0$, so $x_4 = x_6 = 0$ and $x_5 = -x_7$. When $\delta = 1$ this is zero; when $\delta = 0$ the sole possible root line is

$$\langle e_5 - e_7 \rangle. \tag{13}$$

Its ordinary reflection is the transposition (57), so its negative reflection already belongs to H. This contradicts the independence of s_5 from the original end. No such end can extend to a rank-six marking of G.

Theorem 4. *For every odd $q \geq 5$, a string C-group marking of $\Omega_5(q)$ has rank at most five.*

Proof. The endpoint and middle arguments exclude rank greater than six. Suppose that rank six occurs. If every mark is a negative reflection, the acquired reflection bound gives a contradiction. Otherwise at least one middle mark is a halfturn, since all four end marks are negative reflections. The original end $H = \langle s_0, \ldots, s_4 \rangle$ contains that halfturn. Its normal reflection subgroup K cannot be classical by Lemma 2. If it is an ordinary finite alternative, Lemma 3 makes all the end marks negative, again a contradiction. Its only remaining possibility is the deleted S_7 case, excluded by (9)–(13). Hence rank six is impossible. □

Inputs, contributions and present status. The full finite reflection classification and the classical group facts are retained external results. The previously reviewed negative-reflection bound, the C-triple argument, and the F006 module and extremal permutation results are acquired proofs. Their complete required interfaces and finite certificates must accompany any

eventual frozen packet. The new argument treats all orthogonal endpoint spectra, changes the middle marks only to obtain spanning roots, and studies the normal reflection subgroup of an unchanged end. The separate review agent contributed independent working calculations for the nonsemisimple centralizers; that contribution must be disclosed in review provenance. This draft has not yet received complete distinct review. Its accompanying exact arithmetic checks support the centralizer formulas and the deleted symmetric obstruction; they do not verify the retained classification or substitute for review of the uniform proof. This file does not itself confer a principal F001 disposition, conjecture credit, or a compression measurement.

References

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